

Design and Development of IOT enabled IV infusion rate monitoring and control device for precision care and portability

Mohammed Arfan
B-Automate India Pvt Ltd
Bangalore, India
mdarfaneec@gmail.com

Srinivasan M
B-Automate India Pvt Ltd
Bangalore, India
srinivasanm329@gmail.com

Adithya Gowda Baragur
B-Automate India Pvt Ltd
Bangalore, India
adithyagowda2008@gmail.com

Vaishnavi Naveen
B-automate India Pvt Ltd
Bangalore, India
vaish.june@gmail.com

Abstract—This paper proposes a drip monitoring and control device that fits on the existing Intravenous (IV) infusion setup. Normally, doctors and nurses use their experience to estimate the time required for a drip bottle to be empty and for setting drip rate. This makes the IV infusion method to be vulnerable to human error and there are also other risks like back flow of the blood and many other if not attended when empty. This paper proposes an IoT (Internet of Things) based monitoring and control platform for IV infusion setup which enables doctors and nurses to monitor as well as control the IV infusion setup wirelessly while keeping the cost low and making the device highly reliable. The device can be divided into 3 units. The first unit houses an IR sensor for optical based drop detection, second unit houses a microcontroller with a 2.4Ghz Wi-Fi module and the other unit is the actuator to regulate drip rate. In addition, the device is tested along with different parameters like accuracy, stability, reliability, and battery capacity. Finally, the data from the IoT device is reliably transmitted to an online server using HTTP POST request (Hyper Text Transfer Protocol). Then this data is stored in a SQL (Structured Query Language) database and visualized on a webpage as well as a mobile app for easy readability. Also, the attendant could remotely control the device to set drip rate, shut or start and monitor the IV therapy.

Keywords— Medical information systems, Internet of Things (IOT), multi-sensor monitoring platform, Internet of medical things (IoMT), Intravenous therapy.

I. INTRODUCTION

One of the fastest ways to deliver fluids and medications to our body is through intravenous therapy. In this therapy an IV bottle is filled with desired medication in the form of fluid and hung up at a higher level than the patient's body, so that gravitational force can work against the cardiovascular pressure in the body. Even though this method is low cost and effective, it has its own limitations like tissue formation in the needle, rolling of the tube on patient's hand, backflow of blood into the bottle due to the vacuum created on the bottle which can compromise with the patient's life. Hence, monitoring the IV bottle becomes a crucial role as it is necessary to change the bottle on time. Generally, the bottle is changed when it is empty, but the

time not only varies for different medication bottles but also depends on different parameters like quantity, systolic-diastolic and veins contraction.

IV therapy is very dynamic in nature, so a robust monitoring and control of the setup is necessary. Backflow of blood into the bottle can happen if there is any failure in monitoring. In severe cases, if the bottle gets empty and it is not monitored for some time, it can cause insertion of air bubbles [1][2] in the IV setup, so critical monitoring is required otherwise the results of this can be deadly. Currently, this monitoring is done by the patient's attendant and/or by nurses. In countries like India, we have a poor nurse to patient ratio. Over the last 20 years, scholars have explored various medical devices which can be connected to the Internet which will give rise to the Internet of Medical Things (IoMT). Over the past 2 decades, many researchers have tried numerous methods and techniques for sensing the drip levels. One common aim of all the researchers was to develop a system which detects IV fluid without being in physical contact with it. The first generation were focused on low cost and low power consumption. Whereas the modern approaches are more towards integrating alerts systems and monitoring dashboard. But some of these features were platform dependent and hence they had limitations with their usability [4].

Drip chamber being the most widely accepted place for Infrared based drop detection, and this is the most widely explored methods for drop detection. For this optical based drop detection, the performance of the sensor can be affected by ambient lighting in the surrounding. This paper will discuss one of the methods to reduce the ambient effect. Few researches have experimented with software, i.e., by using average filters, or by some physical barriers [5]. The current paper discusses the design, fabrication and testing of an IoT based IV drip monitoring and control device, which can be used to precisely monitor and regulate the drip rate and detect malfunctions in the measurement system. The main consideration for this device was towards having physical and electrical isolation between the Infrared sensor and IV fluid which is achieved by housing the sensor inside a 3d printed case which can make this IoT device retrofittable and independent from any type of fluid and bottle used. Detailed design specifications are discussed in the following section.

II. EXISTING INFUSION MONITORING DEVICES

Following are some of the existing devices available in the market. These devices are just monitoring devices with no control or IV regulation feature in it.



Fig.1. DripAssist®: Infusion Rate Monitor

DripAssist is an IV retrofittable infusion monitoring device which is one of the inspirations for the current project. Its reliable, precise and doesn't need any calibration [8].

There is another similar device named Auto Clamp® which also uses infrared sensing like the one proposed in this paper. But this device needs an IV tube which comes along with the device which offers high precision [8].



Fig.2. Drip-Alert device

Drip-Alert is another such device which has audio and visual alarms when there is deviation from the set drip rate [8].

Conclusion that can be drawn is these devices are just limited to drip monitoring, but they do not demonstrate any control feature to regulate IV therapy.

III. DESIGN AND MATERIALS

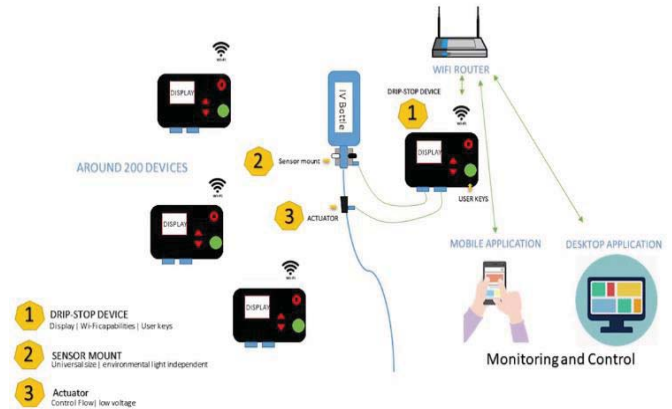


Fig 3: Concept Overview of proposed solution

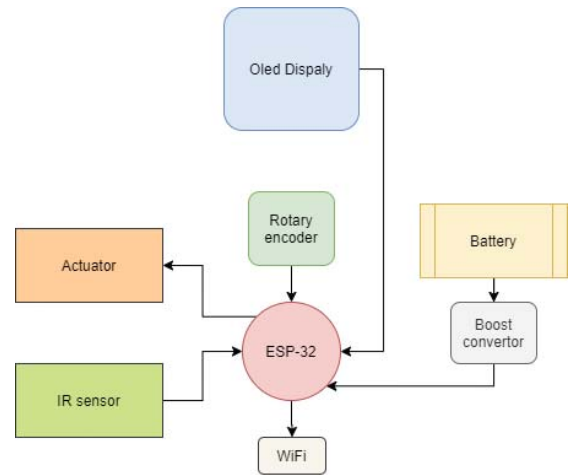


Fig 4: Functional Block Diagram

A. Component Selection, Design and Working.

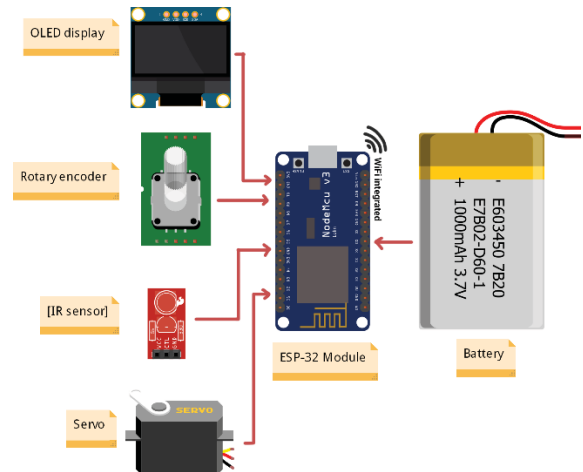


Fig 5: System Setup

1) *Selection of Microcontroller:* ESP32 is a 2-core microcontroller with inbuilt Wi-Fi capabilities and dual-mode Bluetooth with all the features of regular microcontrollers like I2C, interrupts, timers, etc. And with a clock speed of 240MHZ, this becomes an ideal

microcontroller enough to read the data fast enough from the digital pins and to send it over the network.

2) *OLED Display:* SD1306 OLED (organic light-emitting diode) a mono-color (BLUE), 0.96-inch display with 128×64 pixels, this display comes with an I2C and SPI communication protocol which makes it very easy to integrate with ESP32. Adafruit has an excellent library to control this OLED module. One Adafruit_SSD1306.h for the basic interface, while Adafruit_GFX.h is for some advanced graphics.

The drawback of this module was it was slow to write the data on the screen. It would be fine if the display is updated occasionally, but it starts flickering when the frequency of the update is high and the data to be displayed on the single frame was more.

To solve this, instead of rewriting the whole screen again and again only a part of the display was updated by drawing a filled rectangle around the coordinates where data should be updated and then rewriting the updated value. This was way faster than the whole frame update. Therefore, by doing this, the flicker was removed and keeps up with the data from MCU.

3) *Rotary Encoder:* Building a menu system essentially requires 3 buttons minimum to navigate around the screen and to select the menu item.

Therefore, we replaced the whole thing with a single rotary encoder which can rotate both sides infinitely and when it is pushed it acts as a momentary push button. The value from the encoder is decoded in the MCU where '10' or '01' from pin CLK and DT is taken as clockwise turn and '11' or '00' from the pin CLK and DT is taken as anticlockwise turn. This helps in making the menu system that cycles through the menu by rotating the encoder and by clicking the head of the rotary encoder selection is made.

4) *IR Sensors:* IR sensor is used to detect the drip. The emitter and receiver IR LEDs have to be placed inline which is done with the help of a 3d printed sensor mount that has holes to place the emitter on one side and receiver on the other, in such a way that they both face each other. The availability of large variety of IV sets in the market [6] makes the design approach complex because the objective is to come up with a one design that fits all.



Fig 6: 3D printed sensor mount

When the circuit is turned on, we should see the output going high and when this is obstructed by an obstacle the output of the module should go low. But this wasn't observed when the obstacle was water or transparent liquid drop or a translucent body, the IR light from the emitter still passed through these obstacles and hit the receiver, which

was difficult to track. One of the reasons was that the receiver and the emitter had wider transmission and reception thus receiving the light and spreading light which affected the reading.

Therefore, by narrowing the output of the emitter and receiver and by reducing the sensitivity of the receiver to receive only strong IR, it was easier to detect even water droplets with less distortion and at high speed.

Now, when the transmitter IR hits the translucent body, the light disperses, and this intensity of the IR wouldn't be able to trigger the receiver. This narrowing was done by narrow-band IR emitter and heat shrinks, Where the heat shrinks were cut down into small pieces to fit the led and heated up such that all the part of the LED is enclosed except for a small portion on the top. This was done to both the emitter and the receiver.

5) *Actuator:*

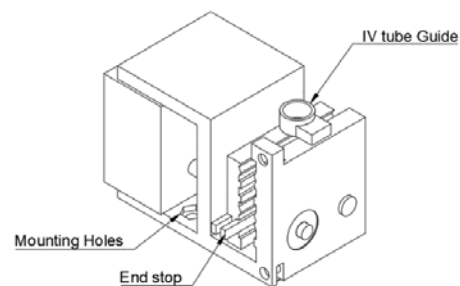


Fig 7: Custom Actuator for IV regulation (Home View)

The need for control over IoT was required to reduce the risk and complication involved in the IV setup. Where the doctor or the nurse can control the fluid rate, start or stop the system as they wish with a web or mobile app. So, we have come up with an actuation system which can reliably stop and release the fluid flowing through the system in a non-invasive way. Since the device is battery operated, the power consumption of the system is also taken into consideration while designing the actuator. The idea was to design an actuator which needs minimum or no holding current unlike other inductive actuators used for similar purpose. As a result, the actuator designed is less prone to accidental release of fluid or power consumption when the actuator is not active.

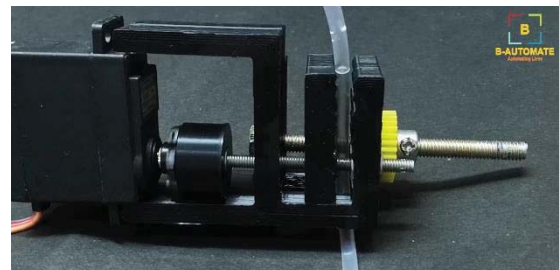


Fig 8: Initial 3D printed custom Actuator

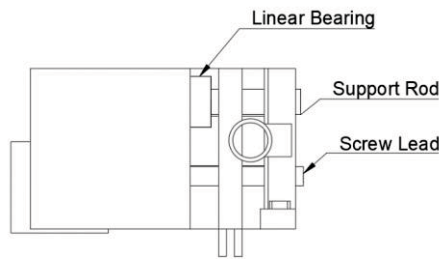


Fig 9: Custom Actuator (Top View)

The actuator is driven by screw lead mechanism, where the screw is controlled by a servo and for stability a support rod is introduced. Top view in fig 9 shows the screw lead mechanism. A movable jaw shown in fig 10 if fixed with a thread is placed in the screw mechanism which moves along when the screw turns in clockwise/anticlockwise direction. This movable jaw applies even pressure to the IV tube which in turn stops the fluids. Drip rate is adjusted by using PID (proportional integral and derivative) algorithm so the set drip rate for the fluid is adjusted by the actuator at equal interval of time. To get the position of the movable jaw, the actuator contains an end stop which is directly connected to the Esp32. With this the microcontroller will know the location of the movable jaw.

This complex mechanism was enclosed in a case shown in fig 11 which was more user friendly.

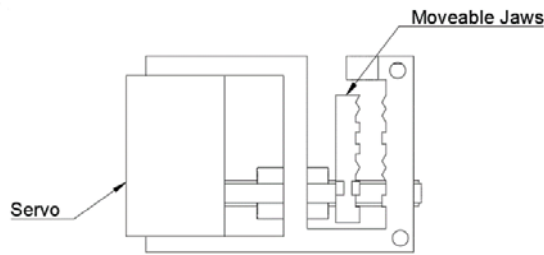


Fig 10: Custom Actuator (Side View)

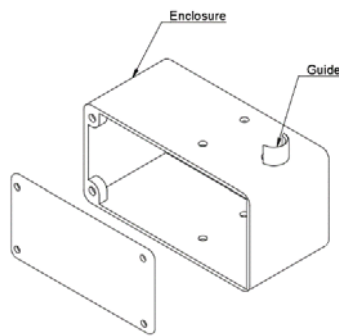


Fig 11: 3D printed Enclosure for Custom Actuator

The problem faced with the servo was the inrush starting current because of the DC motor inside the servo, which draws more current than the battery can supply. To solve this a suitable capacitor is used between the terminal of the servo which helps smooth out the inrush current caused by the DC motor.

IV. ALGORITHM

The algorithm is classic math, to find the unknown by cross multiplying the given value. If 1 drop takes X seconds to drop, then in 1 minute will have $60/X$ drops. But putting this math in the microcontroller was trivial, because the

OLED screen must be updated, and the buzzer has to play accordingly. The answer was to use an external interrupt for counting of drops. In ESP32 all GPIO pins can be used as an external interrupt.

When the sensor is triggered it activates an external interrupt which triggers the ISR (Interrupt service routine) for the falling edge of the sensor value. This ISR sets a flag which is then checked in the main loop. The droplet shape greatly affects the reading as it introduces noise because of the shape which causes the debouncing effect of the interrupt pin and since ESP32 has very high clock speed the reading speed from the interrupt pin is also very fast which leads to wrong counting of drops. This can be fixed by fixing a threshold time limit to take in the next interrupt. If the next interrupt is too fast, then the interrupt ignores the input and takes the input into account only when the interrupt is above the threshold. Since we are using the interrupt, the code is very minimal in ISR (interrupt service routine) so the next interrupt won't affect the ISR. But when the droplet is too fast even before the calculation the ISR is triggered. To avoid this the interrupt is turned off before the calculation and turned back on after the calculation. This might introduce error in the drip rate, but this would be a problem only at a higher rate which would be never used in practice.

This device must read the data from the sensor and simultaneously send data over the internet. The time taken to send the data over the internet is not constant. It varies from time to time and it depends on the broadband speed of the user.

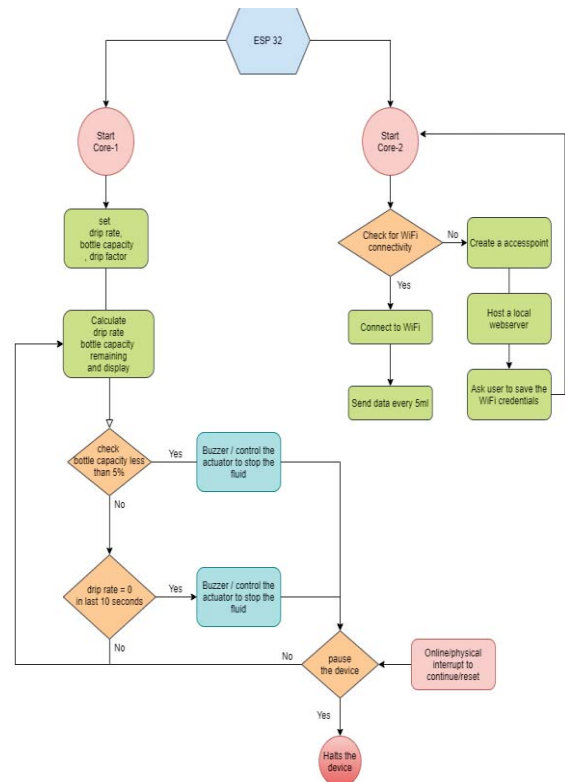


Fig 12: Typical workflow of the system

To resolve this, we used the FreeRTOS feature task handle wherein we divided the main loop into two smaller loops in which one handles the display and the sensor, and the other loop just updates the data over Wi-Fi. This can also be achieved by using multicore features of the ESP32

but the higher clock speed of the MCU made it possible to read the value and display along with sending data over the internet with just one core without any compromise to the accuracy.

V. BATTERY AND CHARGING UNIT

A. Selecting Battery Type

Another important decision to make is that of the battery type and capacity that the circuit runs on. This can be decided based on the power consumed by the circuit, the lifetime of the battery between charges. Once these factors are taken into consideration and are calculated we can decide on the battery capacity and the circuitry necessary to charge it. In this case, since the minimum voltage requirement of most components fall in the regulated voltage range of 3.3v ~ 3.7v, a Lithium-polymer based battery is used.

B. Calculating Battery Specifications

The battery is chosen after some calculations as mentioned before. Since the power consumption of each component varies, upon testing it with a ESP32 Wi-Fi chipset, here are the results.

The ESP32 in full capacity consumes about 140mA of current. It works on a voltage of 3.3v and since the nominal voltage of a single Lithium-polymer cell is 3.7v, a battery operating on the same voltage is chosen.

1) *Equations:* The unit which is used for calculating battery capacity is amps-hour (Ah), which refers to the current output of the battery in amperes for one hour. There is also the milliamps-hour (mAh) unit for ease of calculation.

$$C = I * t \quad (1)$$

where, C is the battery capacity, I is the current in amperes and t is the time in hours.

Upon considering the battery to run approximately for 24 hours in a cycle of charge,

$$C = 140m * 24$$

$$C = 3360mAh$$

The nearest round figured battery capacity available in the market is 3500mAh and hence this is the battery chosen to power the circuit.

Equation (1) is the general equation for calculating the battery capacity of any type of battery defined using the amps-hour unit [3].

C. Charging the Battery

1) *Circuit Schematic:* A schematic of the circuit is illustrated below.

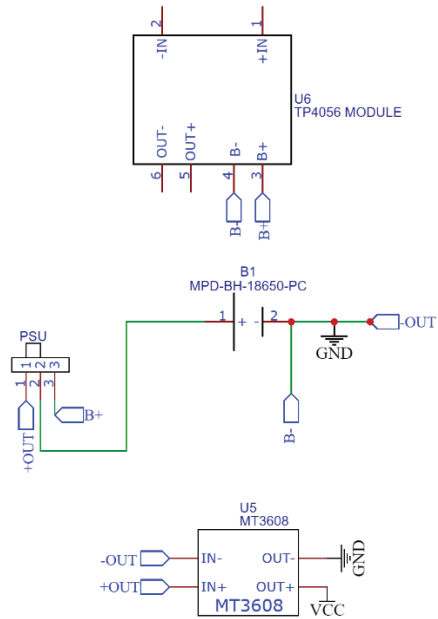


Fig 13: Schematic of battery charger

A 3.7v lithium ion cell is used to power the circuit as well as a BMS (Battery management system) circuit. Since most of the components work on 5v DC, it is necessary to incorporate a regulating circuit to get a stable output. Hence a switching regulator is used to maintain a stable 5v output despite the input voltage provided. Once the required voltage is received, it is either used to charge the battery or as direct output depending on the switch state.

2) *Charging Module (TP4056):* Batteries cannot be charged and discharged at the same time since it violates the rule of Occam's razor; a slide switch is used to choose between charging the battery and powering the rest of the components.

The module used to charge the battery is the TP4056, a typical IC used to charge any Li-ion based battery. The TP4056 module is used because of its onboard battery protection IC. This is an easy to use module due to its current protection and can be used to power a single 3.7v Li-ion cell or higher cells which do not contain their own battery protection. Furthermore, this module prevents the battery from charging at ultra-low voltages (less than 2.4v) and at over-voltages as well as reverse polarity. The IC uses linear charging technique for Li+ battery charging, the advantage of which include low noise, simplicity and low cost.

D. Boosting Circuitry

3) *MT3608 Boost Converter:* Once the battery is charged it is crucial that it supplies the power to the circuitry with the required voltage. The output of the battery is less than that of the needs of the circuit and hence a boosting circuit is used. The module MT3608 is a DC-DC Step Up boost converter.

This module can boost from 5v up to 28v from an input range given anywhere in-between 2v ~ 24v. It has a high

switching speed along with a wide range of current outputs depending on the voltage range used.

VI. WEB AND MOBILE UI

The IOT dashboard is created for the user to monitor, regulate the flow, stop and start the IV therapy. Also keep track of remaining fluid to be dispensed, battery of the device, current drip rate and the drop factor.

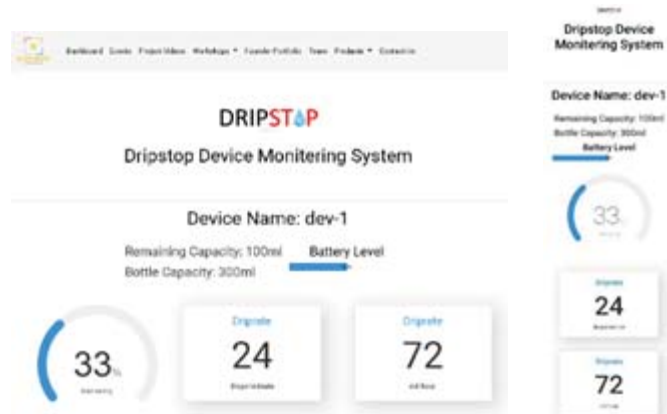


Fig 14: WEB and Mobile dashboard to monitor and control the IV therapy remotely over internet

When the user is accessing the dashboard for the first time, the user will be directed to the registration page wherein his credentials will be registered to MYSQL database. Next, the user will login with the same credentials. Then, the user must add the device to the database. Finally, the user will be directed to a dashboard where all the device data is displayed.

The dashboard is optimized for both Desktop and Mobile browsers. Also, an Android application is available to access

the dashboard. The Android application has been developed in Android Studio using Java programming language.

VII. RESULTS

Multiple prototypes were designed and tested during the entire course of the project finally final 3D printed engineering prototype has been developed shown in Fig 15 and tested under both laboratory and real time conditions. The device demonstrates an accuracy of about ~1ml (which means the actuator is triggered and it squeezes the IV tube to stop the flow when there is 1-2ml still remaining).



Fig 15: Complete System setup during laboratory testing

Also the response time when the device is operated from the Web or mobile interface over the internet is around 2 seconds (Which means the user can start, stop or update drip rate and it will be reflected at the device end in less than 2 seconds). All these is achieved without having to be near the device.



Fig 16: Controlling the device over internet on mobile phone

The device has been tested on multiple infusion sets available in the market, the designed sensor mounts has been tested to fit a wide range of available IV sets.

The battery is estimated to last about a day on a single charge (i.e. typically around 6 500ml bottles spanning around 3hrs each).

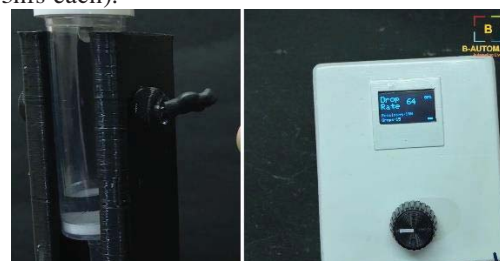


Fig 17: Final 3D Printed Engineering prototype

VIII. CONCLUSION

IoT based monitoring and control platform for IV infusion setup is proposed in this research work. proposed work reduces the human efforts and

requirements in monitoring the infusion setup and enables wireless monitoring system. IR sensor are used to observe the drop detection and drip rate is regulated through actuator unit and the overall process is controlled and executed through a microcontroller unit. Proposed model is highly reliable and it moves the IV therapy into next level utilizing the recent technology.

ACKNOWLEDGMENT

We thank our team members and mentors at B-Automate India Pvt. Ltd. for being compatible and helping us complete this project with their perpetual enthusiasm and hard work as well as our friends and family for their constant support.

REFERENCES

- [1] Pranshu Sardana, Mohit Kalra, Amit Sardana. "Design, Fabrication, and Testing of an Internet Connected Intravenous Drip Monitoring Device", Journal of Sensor and Actuator Networks, 2018
- [2] www.mdpi.com (internet source)
- [3] David Brown "Battery Charging Options for Portable Products"
- [4] Intravenous Stand Design, Mark Vignali, Min Du, Hui Chen, Project report, Huazhong University of Science and Technology, China, August 2006
- [5] Monitoring of Intravenous Drip Rate - V V Kamble, P C Pandey, C P Gadgil, and D S Chaudhary, ICBME, December 2001
- [6] The Drip clip: a lifesaving low cost technology (<https://www.indiegogo.com/projects/the-drip-clip-a-life-saving-low-cost-technology/#/>), Retrieved on 11/11/2015.
- [7] Principles of intravenous infusions and blood transfusions, S Harvey, (<http://shswebpace.swan.ac.uk/HNHarveys/module%20iv%20and%20blood%20transfusions.ppt>), retrieved on 11/11/2015.
- [8] Mohammed Arfan, Lavanya R, 2020, Intravenous (IV) Drip Rate Controlling and Monitoring for Risk-Free IV Delivery, INTERNATIONAL JOURNAL OF ENGINEERING RESEARCH & TECHNOLOGY (IJERT) Volume 09, Issue 09 (September 2020)
- [9] Air Embolism, Risk prevention in infusion therapy, B Braun, (http://francais.bbraun.ch/documents/Knowledge/Content_Wissen_Risikoprävention_Infusionstherapie_Air_EMBOLISM.pdf), retrieved on 11/11/2015.
- [10] A Warning System Based on the RFID Technology for Running Out of Injection Fluid, C Fang Huang, J H Lin, 33rd Annual International Conference of the IEEE EMBS, August-September 2011.
- [11] A CMOS Liquid Level to Frequency Converter with Calibration Circuits for Detecting Liquid Level of Intravenous Drip, C T Chiang, P C T sai, IEEE International Conference on Mechatronics and Automation, August 2014
- [12] J. Wang, B. Li, and D. Qian, "Infusion Monitoring System Based on Wireless Transmission.", IEEE International Symposium on Microwave, Antenna, propagation, and EMC Technologies for Wireless Communications, 2012
- [13] X. Wen, "Design of medical infusion monitor and protection system based on wireless communication technology," in Proc. 2nd Int. Symp. Intell. Inf. Technol. Appl., 2008, pp. 755–759
- [14] Valanarasu, Mr R. "Smart and secure IoT and AI integration framework for hospital environment." J. ISMAC 1, no. 03 (2019): 172-179.
- [15] Haoxiang, Wang. "Trust management of communication architectures of internet of things." Journal of trends in Computer Science and Smart technology (TCSST) 1, no. 02 (2019): 121-131